Narrative audio with local models

BTP progress report
Hari Lakshman | 23CS10009
Presentation: 2 November 2026
Experimental evidence through 28 October 2026

Abstract

This project investigates category-specific audio adaptation and temporal composition as an extension to BackgroundMellow. A local Stable Audio Open pipeline was deployed on a Slurm-allocated NVIDIA V100. A dog-sound LoRA was trained using 24 training clips and selected using eight validation clips. The 200-step pilot reduced fixed-noise validation loss from 0.5088 to 0.2878, while the listener reported mostly similar sound quality and a modest improvement in one sample. Two temporal composition experiments passed implementation checks but did not demonstrate reliable bark placement. In the regional-conditioning experiment, all 12 regional outputs lacked audible barking according to the listener.

A separate-stem story baseline has therefore been implemented for a complete narration, music and effects demonstration. It combines CPU text-to-speech, generated sound candidates, measured sentence boundaries and deterministic waveform placement. Job 30111 completed a 62.33-second narrated mix, which the listener described as almost perfect overall, with some effects entering early or during speech. This report separates verified experimental results from newly implemented capabilities and proposed follow-up work.

Current status

Verified	Remaining refinement
Local generation; dog LoRA; two negative timing studies; English narration and a 62.33-second story mix	Precise cue timing; second story; broader listening evaluation; local Whisper word-timing integration awaits server validation

Main conclusion: adapter training is feasible within the available hardware, but lower denoising loss and correct mask arithmetic have not established successful temporal sound composition.

1. Background and research objective

BackgroundMellow, by Ajitesh Jamulkar and Aritra Hazra [1], treats narrative soundscape generation as orchestration and signal processing. The supplied paper describes specialist generation or retrieval, a TTS master track, word-level timing, DSPred parameter prediction and waveform superimposition. Separate-stem mixing is therefore a baseline inherited from this problem setting, not a new contribution claimed here.

The extension investigates low-rank adaptation of the audio denoiser and composition of its predictions over time. The working name C-MoLo denotes this proposed category-specialist approach; novelty and superiority have not been established. Only a dog adapter has been trained. Rain, piano and the other demo categories use the pretrained base model.

Research questions

RQ1: Can a small category dataset support stable, low-cost audio adaptation?
RQ2: Does adaptation improve perceived category quality beyond prompt selection?
RQ3: Can temporal prediction masks place events while preserving background audio?
RQ4: Can a transparent local baseline produce a usable narrated story with inspectable layers?

Architecture and boundaries

Stage	Separate-stem demo	Experimental latent branch
Input	Authored story scenes and categories	Scene prompt(s) and target interval
Narration / context	MMS English TTS per sentence	Text embeddings and full-clip duration
Sound generation	Stable Audio candidates; dog LoRA optional	Same latent through base and expert
Timing	Measured sentence lengths; authored offsets	Soft codec-frame mask
Combination	Place stems, crossfade beds, duck under speech	Blend v predictions; one scheduler update
Output	Mix WAV, narration, stems and metadata	Single decoded soundscape and metadata

The story implementation is currently CLI-first. Its plan is authored rather than LLM-generated, and it does not perform word-level forced alignment. A server and frontend are deferred until the audio workflow is stable.

2. Reproducible experimental setup

Item	Configuration
Backbone	stabilityai/stable-audio-open-1.0 [2]
Pinned revision	f21265c1e2710b3bd2386596943f0007f55f802e
Runtime	Python 3.11.17; torch 2.7.1+cu118; Diffusers 0.32.2; Transformers 4.49.0
Compute	Slurm gpupart_v100; Tesla V100-PCIE-32GB; shared NFS repository
Inference	100 steps; guidance 7.0; FP16; cosine DPM scheduler; v_prediction
Dataset	ESC-50 dog class [3]: 40 clips, five seconds each
Split	Folds 1-3: 24 train; fold 4: 8 validation; fold 5: 8 test

The user accepted the 40 recordings after listening, noting some reflections and aggressive barking. Source IDs and exact hashes were checked across splits. Near-duplicates and possible overlap with the base model pretraining data are not excluded. Test recordings were excluded from latent caching, optimization and checkpoint selection, although the user had heard them during review.

Adapter and objective

The backbone was frozen. Rank-8, alpha-8 adapters were attached to self- and cross-attention query/value projections: 96 projections and 2,064,384 trainable parameters. Training used AdamW, learning rate 1e-4, zero weight decay, batch size one with four gradient-accumulation steps, norm clipping at one, and 10% text-conditioning dropout. Adapter arithmetic and optimizer state were FP32; the frozen transformer used FP16 and activation checkpointing.

For a clean codec latent x, Gaussian noise n and normalized time t, alpha = cos(pi t/2), beta = sin(pi t/2), noisy input = alpha x + beta n, and target velocity = alpha n - beta x. This is compatible with the pinned scheduler parameterization. It is not a reproduction of original pretraining: codec posterior means were cached, audio was padded to model length, and active/padded frames received weights 1/0.1.

Validation used eight clips at fixed noise levels 0.2, 0.5 and 0.8. The checkpoint with the lowest validation loss was selected. The two-step smoke run used a smaller validation subset and should not be numerically compared with the full pilot.

3. Results: generation and dog adaptation

Job	Result	Interpretation
29955	Five-second base dog generation completed	User heard clear barking; possible birds
29956	Six samples: two prompts x three seeds	Prompt wording is a necessary control
29962	Two-step LoRA smoke passed	Gradients, zero-adapter identity and reload tested
29967	200-step training; 321.15 seconds	Best checkpoint at step 200
29974	Six paired LoRA samples completed	Mostly similar quality; one modest improvement

Step	Validation loss
0	0.5087598581
50	0.3079632645
100	0.2956833843
150	0.2903111530
200	0.2877913692

The relative validation-loss reduction was 43.43%. Reloaded adapter predictions matched exactly in the reported test. The listening comparison used the same model revision, prompts, seeds and generation settings. The listener found similar bark quality overall and a small improvement in the previously reported bird-chirping sample.

Interpretation: the pilot establishes that the adapter can be trained and used for inference. It does not establish a 43.43% perceptual improvement, generalization to unseen acoustic conditions, or superiority over the base model. The comparison was informal and unblinded, with one listener and six samples. Outputs were not loudness-matched; differences in level are a possible confound.

4. Results: temporal composition

At each denoising iteration, both branches receive the same latent and timestep. Their velocity predictions are combined as v_mix = v_background + W(tau) [v_event - v_background], then the pipeline applies classifier-free guidance and one scheduler update. Masks use the actual codec spacing, 2048/44100 seconds per frame. Attention and decoding can spread effects beyond the nominal mask.

Experiment	Design	Listener result
Adapter-only masking Job 30033	Base, full adapter and mask at 3-6 seconds; three seeds	Seed 42: full adapter barked continuously, masked output had only rain. Seed 123: rain only. Seed 2026: existing bark times largely unchanged.
Regional conditioning Job 30101	Rain-only background; rain-plus-dog event. Windows 1-4 and 6-9 seconds; adapter off/on; three seeds	No audible barking in all 12 regional samples. Only global base/LoRA at seed 2026 barked, around 2, 3 and 6 seconds. Seed 123 was very quiet.

The temporal and regional endpoint checks reproduced their zero-mask and full-mask references exactly. All copied WAV hashes and formats were verified. Job 30033 completed its nine comparison files in 160.52 seconds including loading; job 30101 completed its 21 files and endpoint checks in 461.48 seconds. Regional inference used two model passes per denoising step, approximately doubling generation time in these runs.

Acoustic conclusion: neither experiment demonstrated temporal bark placement. Absence of barking outside the interval cannot count as success when barking is also absent inside it. The first temporal study preserved acceptable rain and had no reported distortion; no additional distortion rating was inferred for the regional run.

Possible mechanisms include competition between regional predictions and nonlocal model behavior, but the experiments do not identify a cause. Further changes require controlled ablations. Claims that latent blending inherently prevents phase cancellation, guarantees simultaneous events or improves sound quality are not supported by these results.

5. Story demo and evaluation plan

The new baseline uses the eight-sentence story "The light in the forest". It covers rain, forest ambience, trees, wind, walking footsteps, running, dog barking, creaking wood, a creaking door and gentle piano. MMS English TTS [4] runs locally on CPU. Sentence waveforms are resampled to 44.1 kHz, separated by 2.5-second gaps, and measured to create the narration timeline.

Stable Audio generates two candidates per category. The dog adapter is enabled only for dog clips. Candidate selection is explicit in a JSON file; candidate zero is provisional until listening. Ambience and piano can be extended with crossfaded loops. Isolated effects play once and are padded with silence if necessary. The mixer applies fades, narration-driven background ducking and sample-peak headroom. This is not LUFS normalization or true-peak limiting.

The new code passed local tests, including cue coverage, looping, level limits and narration mixing; the full local suite contains 28 passing tests. Job 30111 completed all 20 sound candidates and the first narrated mix. All 16 mix/narration/stem hashes and all 20 candidate hashes matched locally. The final mix had no full-scale PCM samples. Its 1.73-second render time excludes TTS and sound generation. The listener reported some early effects; the opening rain intentionally precedes narration by one second. A new CPU Whisper word-timing refinement has been implemented but is not yet server-validated. The sound model is an initial option for a short piano bed, not a guarantee of coherent long-form music. If a sound candidate fails, it must be replaced or regenerated rather than counted as present.

Evaluation to complete before the presentation

Measure	Procedure
Event coverage	Count intended categories with audible correct sounds; report missing events
Timing	Annotate audible onset/offset relative to authored cue windows
Narration	Listen for intelligibility, pronunciation and masking by effects
Music / ambience	Check levels, loop seams and continuity
Reproducibility	Keep model revisions, seeds, selected filenames, hashes and run logs

Add a second story and retain every candidate choice. Use the separate-stem baseline as a reproducible demo and a comparison point, not as evidence that latent composition succeeded. Future research can investigate region-specific conditioning, duration-aware event training and larger specialist datasets. Frontend work should not displace audio validation before 2 November.

6. Delivery schedule and references

Date	Planned outcome
28-29 October	Refine timing and select the better story mix
30 October	Second story and listening comparison
31 October	Freeze examples and update report/slides
1 November	Rehearse; save offline audio and presentation backups
2 November	BTP presentation/submission deadline

References

[1] Ajitesh Jamulkar and Aritra Hazra. BackgroundMellow: A Multi-Modal Cohesive Framework for Narrative-Driven Rich Cinematic Soundscape Generation. arXiv:2607.11364v1, supplied manuscript, 13 July 2026.

[2] Stability AI. Stable Audio Open 1.0 model card. https://huggingface.co/stabilityai/stable-audio-open-1.0 . Accessed 28 October 2026. Model license: Stable Audio Community License; retain downloaded license materials.

[3] Karol J. Piczak. ESC-50 dataset repository. https://github.com/karolpiczak/ESC-50 . Dataset provenance and split metadata retained during preparation. Verify source/subset licensing before redistribution.

[4] Meta. MMS English TTS model card. https://huggingface.co/facebook/mms-tts-eng . Accessed 28 October 2026. Model license: CC-BY-NC-4.0.

[5] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685. Used as the adaptation principle; this project uses a custom audio implementation.

[6] Hugging Face Diffusers 0.32.2, StableAudioPipeline and CosineDPMSolverMultistepScheduler source. Exact installed source inspected for conditioning, velocity parameterization and scheduler integration.

Evidence index

Training: job 29967 logs. Prompt and LoRA comparisons: outputs/prompts-29956 and outputs/lora-prompts-29974. Temporal study: outputs/temporal-30033. Regional study: outputs/regional-30101. Listening observations are recorded in docs/RESULTS_2026-10-26.md, docs/TEMPORAL_PILOT.md and docs/REGIONAL_PILOT.md. Story evidence: outputs/story-demo-30111 and outputs/sound-library-30111. Demo instructions are in docs/STORY_DEMO.md; timing refinement is documented in docs/ALIGNMENT_DEMO.md.

Evidence boundary: no formal blind human evaluation, automatic event-detection score, FAD or CLAP result is claimed. The narrated demo is verified as an engineering run with positive informal listener feedback; word-level timing accuracy and category coverage are not formally scored.